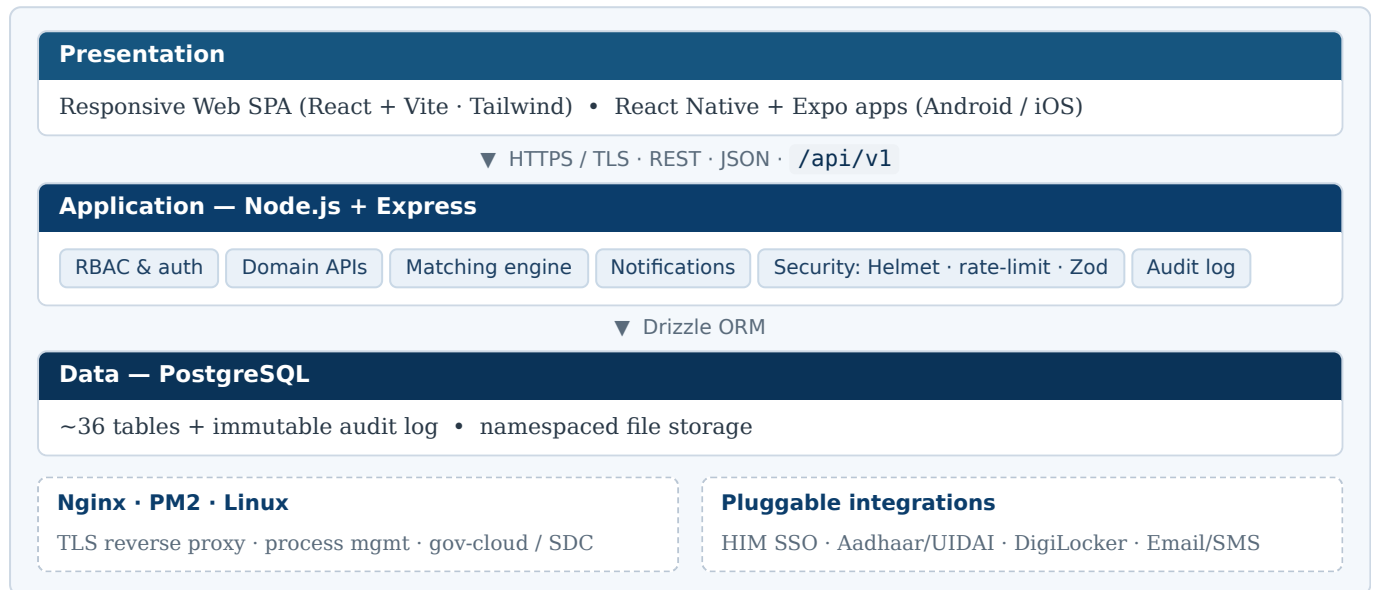


HireStream — Technology Stack

Overseas Placement Portal & Mobile Application · HPSEDC

HireStream — the HPSEDC Overseas Placement Portal & Mobile Application — is a secure, bilingual, cloud-native platform built on a modern, proven, open-source technology stack. This sheet summarises the technologies used and how they fit together.

SYSTEM ARCHITECTURE — THREE LOGICAL TIERS



Web front-end

Area	Technology
Core	React 18 + Vite + TypeScript
Routing & data	Wouter (routing), TanStack Query (server-state & caching)
Forms & validation	React Hook Form + Zod (schema validation)
UI & styling	Radix UI / shadcn components, Tailwind CSS , Recharts (charts)
Localisation	i18next — bilingual English + Hindi

Back-end & API

Area	Technology
Runtime	Node.js + Express + TypeScript
API	Versioned REST API (/api/v1), JSON over HTTPS
Data access	Drizzle ORM + drizzle-zod (schema-derived validation)
Background jobs	node-cron (digests, SLA checks), Winston (structured logging)

Database & storage

Area	Technology
Database	PostgreSQL — ~36 relational tables + an immutable audit log
Files	Namespaced document/photo storage; uploads verified by file type & magic-bytes (PDF / JPG / PNG)

Security

Area	Technology
Access control	Role-Based Access Control (Candidate / Agency / Employer / Admin)
Authentication	passport + express-session (PostgreSQL-backed, rolling idle timeout), OTP (otplib), JWT for mobile, bcrypt password hashing
Hardening	Helmet security headers, rate limiting, input validation, encrypted integration credentials, full audit trail

Mobile application

Area	Technology
Apps	React Native + Expo — Android & iOS, consuming the same REST API
Features	Mobile auth (refresh tokens), push notifications, offline-tolerant config

Documents & communications

Area	Technology
Generation	PDFKit (PDF), QRCode, Archiver (bulk ZIP export)
Messaging	In-app notifications, Nodemailer (email), pluggable SMS adapters

External integrations — pluggable & admin-configurable

HIM Access SSO · Aadhaar / UIDAI · DigiLocker · Email & SMS gateways — all designed as adapters and activated from the admin console once production credentials are provided. No code change required to switch providers.

Infrastructure & deployment

Area	Technology
Hosting	Linux · Nginx (TLS reverse proxy) · PM2 process management
Cloud	Deployable to government cloud / State Data Centre ; stateless API scales horizontally
Targets	Page load < 3s · 5,000 concurrent users · 99.9% uptime · data backups + health monitoring

Standards & compliance

- **GIGW** — accessibility for *divyangjans*, State-Government header uniformity
- **HTTPS / TLS** encryption in transit; sensitive data encrypted at rest
- **ISO 27001**-aligned security practices; GDPR / PDPA-equivalent data protection

Why this stack

- **Modern & proven** — industry-standard, actively maintained open-source technologies.
- **Secure by design** — RBAC, encryption, hardened sessions, and complete audit trails.
- **Scalable & cloud-ready** — stateless API and horizontal scaling for growth.
- **Maintainable** — TypeScript end-to-end, modular design, documented for handover.